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Tianjin Action Plan for Adapting to Climate Change

In order to thoroughly implement Xi Jinping's ecological civilization thought and fully implement the Decision-making and deployment, implementation of the national strategy for actively responding to climate change, and strengthening climate adaptation to effectively prevent the adverse impacts and risks of climate change, in accordance with the This plan is formulated in accordance with the requirements of the "Strategy for Adapting to Climate Change 2035".

1. The Situation We Face

The trend of global warming is intensifying, and extreme weather and climate events and meteorological disasters are occurring. The frequency shows a clear increasing trend and has become a major challenge facing human survival and development. Tianjin is located in the mid-latitude monsoon climate zone, in the northeast of the North China Plain, in the Haihe River Basin. The lower reaches, bordering the Bohai Sea to the east and the Yanshan Mountains to the north, are sensitive and vulnerable areas to the impact of climate change. In the context of global warming, the temperature in Tianjin has increased significantly since 1961. The warming rate is 0.38ÿ/10 years, higher than the national and global averages during the same period. The sea level is fluctuating and rising. From 1980 to 2022, the sea level rise rate in Tianjin is The rate is >4.0 mm/year, which is higher than the sea level rise rate along the coast of China and the global average during the same period. Average level.

As a megacity and port city with a high concentration of population and economic activities, the climate Changes and urbanization effects are intertwined and coupled, affecting urban development and people's production and life. The adverse impacts are gradually increasing. Studies have shown that the warming trend of the climate system will continue. The climate change risks faced by Tianjin in the coming period will be more prominent, with high temperature heat waves, The frequency and intensity of extreme weather and climate events such as heavy rainfall have increased, and water shortages have

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The situation will not improve significantly, sea levels will continue to rise, and the frequency of extreme sea level events will increase.

In the face of the increasingly prominent risks of climate change, it is urgent to carry out the orderly adaptation to climate change.

Take action to enhance urban climate resilience and ensure safe operation and sustainable development of cities.

II. General requirements

1. Guiding ideology

Guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, we will fully implement the Party

The spirit of the 20th National Congress of the Communist Party of China is to coordinate development and security and implement a national strategy to actively respond to climate change.

strategy, adhere to the system concept, attach equal importance to mitigation and adaptation, and take improving the ability to adapt to climate change as the priority.

It is an important part of improving the city's governance capacity and level, and constantly strengthening water resources, natural ecology

systems, oceans and coastal zones, and other natural ecosystems, infrastructure and human settlements, agricultural

industry and food security, health and public safety, sensitive secondary and tertiary industries, and other economic and social systems

Climate resilience, effectively responding to the long-term adverse effects of climate change and reducing extreme weather

Disaster losses caused by climate events help build a beautiful Tianjin where people and nature coexist in harmony

Construction of a socialist modern metropolis.

2. Basic principles

Adhere to prevention first and proactively adapt. Identify the long-term slow-onset impacts and sudden

to reduce the risk of extreme weather and climate events, and to conduct in-depth assessments of the impacts and risks of climate change.

Establish a bottom-line mindset, seek benefits and avoid harm, take the initiative, take positive actions, and strive to prevent and resolve

Addressing major climate risks.

Adhere to systematic advancement and key breakthroughs. Integrate climate change adaptation into economic and social development.

The whole process of high-quality development and high-level protection of the ecological environment, focusing on key vulnerable areas and regions

strengthen climate change adaptation actions, and continuously improve economic and social systems and natural ecosystems.

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System climate resilience.

Adhere to the principle of following nature and adapting scientifically. Practice the concept that green mountains and clear waters are invaluable assets.

Make full use of nature-based solutions and adopt scientific and reasonable climate adaptation measures according to local conditions.

We should take action to take advantage of favorable factors and guard against unfavorable factors to achieve economic development and climate change.

A win-win response.

Adhere to synergy and multi-party linkage. Coordinated promotion of adaptation and mitigation actions, fully

Mobilize the enthusiasm and participation of the whole society to actively adapt to climate change, strengthen work coordination,

Information sharing and innovation support will be strengthened to enhance the synergy of policies and measures and form a

Work together.

3. Main objectives

By 2025, a climate change adaptation work system will be basically established, and scientific prevention and

The ability to respond to extreme weather and climate events, climate disasters, and marine disasters continues to improve.

The city's infrastructure has effectively enhanced its ability to adapt to climate change, and the natural ecosystem has

Climate change capabilities have been further enhanced, and climate change-related publicity and education, scientific and technological support, etc.

The capabilities have been significantly improved, and an atmosphere of social participation in climate change adaptation actions has been initially formed.

By 2030, the climate governance policy system will continue to improve, extreme weather and climate events will be

The monitoring, early warning and disaster prevention and mitigation systems for disasters are basically complete, and key areas and regions are suitable for disasters.

Significant progress has been made in addressing climate change, and the climate of natural ecosystems and economic and social systems has

Climate vulnerability has been significantly reduced, the concept of climate change adaptation has been effectively popularized throughout society, and climate adaptation

The construction of responsive cities has made positive progress.

By 2035, the monitoring, early warning and disaster prevention and mitigation systems for extreme weather and climate events will be

The construction is further improved, the climate risk management and prevention system is basically mature, and the major atmospheric

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Climate-related disaster risks are effectively controlled, and the technical system and standards for adapting to climate change are

system has become more complete, the ability of the whole society to adapt to climate change has been significantly improved, and climate-adaptive cities have

The city was basically built.

III. Strengthening Climate Change Monitoring, Early Warning and Risk Management

4. Improve the climate change observation and monitoring system

1. Improve the climate change observation system.

Based on the principle of "one country, one world", we will build an integrated, coordinated and efficient precision meteorological monitoring system for land, sea, air and space.

Improve the intelligent meteorological observation system in large cities and enhance the precision monitoring capability of extreme and disastrous weather.

Strengthen vertical meteorological observation and enhance the three-dimensional observation capability of land and sea.

Collaborative technology, improve the observation and application capabilities of drones, establish an aerial remote sensing observation system,

Strengthen the capacity building of Tianjin Data and Application Center of High Resolution Earth Observation System.

Build a greenhouse gas observation network covering major ecological functional areas. Strengthen meteorological, water, environmental protection,

Cooperation and data sharing between surveying and mapping departments to carry out multi-layer, multi-disciplinary and mega-city

Stereoscopic observation test. (Municipal Meteorological Bureau, Municipal Water Affairs Bureau, Municipal Ecological Environment Bureau, Municipal Planning Bureau

The Resources Bureau is responsible for the division of responsibilities, and the people's governments of each district are responsible for implementation.

The people's government will implement it and will not list it anymore)

2. Improve the monitoring and analysis capabilities of the climate system. Based on national multi-sphere and multi-source observation

data and basic data products, establish a quantitative monitoring indicator system, and standardize the

Climate change monitoring technology and methods, and carry out research on climate change facts and major weather and climate in Tianjin

Monitor the entire process of the event. Strengthen the monitoring of the climate and environmental effects of human social and economic activities.

Focus on major projects of ecosystem protection and restoration, strengthen

Monitor the impact of climate change on ecological construction. Strengthen the impact of global warming on the coastal areas of our city.

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Monitoring of regional impacts. Building a high-quality and secure meteorological data and information network system.

The Meteorological Bureau and the Municipal Planning and Resources Bureau are responsible according to their respective duties)

5. Strengthening climate disaster prediction and early warning

3. Improve the accuracy of meteorological disaster prediction and forecasting.

Prediction system products to improve the climate prediction level of Tianjin and Haihe River Basin in the next 10-30 years.

Strengthen the study of weather mechanisms, climate laws, climate change, and the mechanisms of disastrous weather.

Comprehensively carry out refined grid forecasting, strengthen artificial intelligence, neural network

The application of technologies such as the Internet of Things in heavy rain forecasting, extreme high/low temperature forecasting, etc., improves the

Improve the objective forecasting capabilities of the energy grid. Improve the monitoring and forecasting of meteorological disasters by disaster type and key industry.

The meteorological disaster monitoring rate has reached 99%.

Regional ozone and aerosol simulation and prediction capabilities. (Municipal Meteorological Bureau)

4. Strengthen early warning for extreme weather and climate events. Establish and improve typhoon, rainstorm and storm

Establish a forecast and warning system for extreme weather such as tidal waves, and dynamically update forecast and warning methods.

Forecasting and warning are integrated with hydrology, geology and other fields.

Event warning information release platform, improve extreme weather and climate events and urban waterlogging,

Small river floods, mountain torrents, geological disasters, forest fires, marine meteorological disasters, etc.

Improve the forecast and warning capabilities of meteorological disaster risks, and improve the scientificity, accuracy and timeliness of warnings.

The public coverage rate of disaster information release reached 95%. (Municipal Meteorological Bureau, Municipal Planning and Resources Bureau,

The Municipal Water Affairs Bureau and the Municipal Emergency Management Bureau are responsible according to their respective duties)

6. Strengthening the assessment of climate change impacts and risks

5. Improve the assessment technology level and basic capabilities. Strengthen the ability to assess meteorological disaster risks.

The focus is on the main weather conditions that affect Tianjin, such as heavy rain, strong winds, high temperatures, and drought.

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To carry out risk assessment and prediction technology research and development, and build an integrated meteorological

Comprehensive disaster risk assessment and decision-making information support system. Conduct regional climate change impact assessment.

and research on quantitative assessment technology of climate risk. Promote the construction of climate change data management system,

Establish a big data sharing platform covering multiple spheres and human activities to achieve climate change and

The data on their socio-economic impacts will be shared.

Combine risk survey and disaster risk assessment, compile natural disaster risk zoning maps and prevention and control zones

(The Municipal Meteorological Bureau and the Municipal Emergency Management Bureau are responsible for the division of responsibilities)

6. Strengthen urban climate risk assessment. Scientifically analyze the current status of climate change in Tianjin.

Identify the medium- and long-term impacts and risks of climate change on urban society, economy and ecology.

Towards urban planning and construction, ecological protection and restoration planning and construction, and explore the promotion of climate change

Impact and risk assessment. Based on long-term series meteorological observation data and meteorological disaster data

Based on the above, we will build a climate risk indicator system and compile urban climate risk indicators at different spatial scales.

Carry out city health check assessment, gradually improve the city health check system, explore the city

Application of physical examination and assessment results in the recognition and management of urban climate risk points.

Environmental Bureau, Municipal Meteorological Bureau, Municipal Housing and Urban-Rural Development Committee, Municipal Planning and Resources Bureau, Municipal Urban

The Management Committee is responsible according to the division of responsibilities)

7. Strengthening comprehensive disaster prevention and mitigation

7. Improve the disaster warning and emergency management system. Strengthen the prevention and control of climate change such as drought and floods.

Study on the evolution mechanism of natural disasters under the influence of climate change, classify and construct disaster prediction and analysis models,

Improve the accuracy of meteorological disaster prediction, analysis and assessment. Incorporate meteorological disaster prevention into Tianjin

The construction of natural disaster prevention and control system focuses on low-frequency but high-impact extreme weather and climate

We will revise and improve the multi-level meteorological disaster emergency plan.

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Based on the principle of land management, improve the urban and rural meteorological disaster prevention and mitigation system and mechanism.

The "Elephant" campaign integrates meteorological disaster prevention and reduction into the urban and rural grid management system.

Meteorological disaster prevention level assessment, and supervision of the implementation of meteorological disaster prevention measures.

The Municipal Bureau of Water Affairs, the Municipal Planning and Resources Bureau, and the Municipal Emergency Management Bureau are responsible for their respective duties.

8. Strengthen disaster emergency response capabilities. Improve natural disaster management including meteorological disasters.

disaster relief emergency plans, strengthen disaster monitoring and early warning, information reporting, emergency response,

Post-disaster rescue and reconstruction, and emergency support system construction. Strengthen various professional

Strengthen the construction of disaster relief teams and disaster management personnel, and strengthen the construction of grassroots emergency rescue forces.

Encourage and support the development of social emergency response forces. Strengthen the reserve of emergency materials and equipment, and enhance the

Socialization of emergency supplies for enterprises, social organizations, streets (townships), communities (villages)

In accordance with the principles of reasonable layout and moderate scale, make full use of schools, public

Public facilities such as parks, squares, and stadiums should be planned and set up in an overall manner to provide emergency shelters.

And set up obvious signs. (The Municipal Emergency Management Bureau and the Municipal Meteorological Bureau are responsible for the division of responsibilities)

IV. Improving the ability of natural ecosystems to adapt to climate change

8. Water Resources

9. Improve the water resources and flood and drought disaster monitoring system.

Improve infrastructure construction, optimize the layout of surface water monitoring sections, and strengthen groundwater user monitoring and metering.

Optimize the layout of the hydrological monitoring station network, upgrade the existing hydrological monitoring stations, and improve the entry

The network of hydrological monitoring stations for rivers entering the sea is supplemented and improved to monitor water level, flow, water quality, etc.

We will strive to achieve full coverage of hydrological monitoring of small and medium-sized rivers in existing flood control tasks.

Real-time online monitoring of water, rainfall, construction conditions, etc., explore new monitoring methods, and vigorously improve

Improve water resources and water affairs intelligent monitoring and perception capabilities. Optimize flood control monitoring sites and establish

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The river system forecast and early warning work mechanism was established, and the construction of flood control emergency information system was carried out.

Bureau)

10. Deepen the intensive and economical use of water resources. Establish a rigid constraint system for water resource use.

We will implement dual control of total water consumption and intensity, strictly follow the development and utilization of water resources and

Water efficiency control red line. Deepen the construction of a water-saving society, carry out national water-saving actions, strengthen

Chemical industry water conservation and emission reduction, agricultural water conservation and efficiency improvement, urban water conservation and loss reduction, promote water-saving enterprises

water-saving industries, water-saving public institutions, water-saving universities, and water-saving residential communities.

By 2025, the total water consumption in the city will be controlled within 3.5 billion cubic meters, and the city's domestic consumption will be

Water consumption per GDP will drop by 10% compared to 2020. By 2035, a water-saving society

The city's total water consumption is controlled within 4.22 billion cubic meters, and water resources are used

Maintain domestic advanced level of efficiency. (Municipal Water Affairs Bureau, Municipal Development and Reform Commission, Municipal Industry

The Information Bureau and the Municipal Agriculture and Rural Affairs Commission are responsible for their respective duties.

11. Enhance urban and rural water supply security capabilities. Further optimize and improve the urban and rural water supply pattern.

Strengthen water resource security along multiple routes, deepen water source protection for the Luan River, and expand the middle route of the South-to-North Water Diversion Project

Water supply scope, promote the high-quality development of the follow-up projects of the South-to-North Water Diversion Project.

Integrative construction will basically realize the integration of urban and rural water supply. Accelerate the promotion of sewage resource utilization,

Actively develop seawater desalination according to local conditions. By 2025, urban and rural water supply security can be guaranteed.

The capacity has been significantly enhanced, the utilization rate of recycled water has reached more than 50%, and the utilization of desalinated seawater has reached

By 2035, we will basically build a high-efficiency and integrated urban and rural

The water supply security system has formed a "three-water sharing, five-reservoir coordination, multi-source mutual assistance,

The new pattern of water supply security guarantee of "urban and rural coordination". (Municipal Water Affairs Bureau, Municipal Development and Reform Commission

The Commission is responsible for the division of responsibilities)

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12. Improve the flood prevention capabilities of the river basin. Strengthen the regulation of river systems and reduce The river channel is shrinking due to siltation, and the river channel's flood discharge capacity is restored. Accelerate the filling of the gaps in flood control infrastructure, Focusing on the construction of flood channels and flood storage areas, we will continue to promote the prevention and control of mountain torrents. Improve water disaster prevention capabilities and super-standard flood response capabilities. The city, district, township (street) flood prevention responsibility system, with the chief executive as the core, Improve emergency response capabilities. By 2025, the compliance rate of first-level flood-carrying river embankments will reach 84%. By 2035, the flood control and disaster reduction capacity will be basically completed.

(Municipal Water Affairs Bureau, Municipal Emergency Management Bureau The bureau is responsible according to the division of responsibilities)

13. Promote the comprehensive management and restoration of key rivers and lakes. Make every effort to build an ecological water network and strengthen Comprehensive management of the "six rivers and five lakes" including the Yongding River and Chaobai River. Promote beautiful rivers and lakes Protect and build a group of beautiful places with rivers, water, fish, grass, and harmony between people and water. Lihe Lake. Continue to promote soil and water loss control. Strictly control groundwater over-exploitation, except for emergency In areas with groundwater overexploitation, zero groundwater exploitation will be basically achieved. By 2025, Under normal water supply conditions, the ecological water volume (water level) of important rivers and lakes is basically guaranteed. The soil and water conservation rate has been raised to 98.55%. By 2035, the water ecological space will be effectively protected. The soil and water conservation rate has increased to 98.8%.

The Planning and Resources Bureau is responsible according to the division of responsibilities)

9. Natural Ecosystems

14. Improve the comprehensive monitoring system of natural ecosystems. Deploy and coordinate the investigation and monitoring of forests, grasslands, wetlands and wild animals and plants. Focusing on key regional watersheds, ecological protection red lines, nature reserves, and important ecological functions

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Conduct ecological status surveys and assessments in the districts and other areas. Conduct natural resource surveys and monitoring in light of Tianjin's situation.

Actively apply for national key technology research pilot projects.

Establish a special natural resources monitoring database and timely submit monitoring data to the national main database.

Establish a city-wide land "three surveys" database management system and a shared service platform to promote land

Application and sharing of "three surveys" data. Carry out meteorological monitoring and evaluation of ecological quality,

Monitoring and evaluation of climate carrying capacity, meteorological impact assessment of changes in ecological service functions, and

Ecosystem meteorological impact pre-assessment and risk warning. (Municipal Planning and Resources Bureau, Municipal Ecological

The Environmental Bureau and the Municipal Meteorological Bureau are responsible for the division of responsibilities)

15. Improve the natural ecosystem protection and supervision system. Integrate and optimize various natural

Protected areas and improve the carrying capacity of natural ecological space. Strengthen ecological protection supervision and management,

Continue to carry out the "Green Shield" nature reserve to strengthen supervision of special actions, carry out ecological protection red

Remote sensing monitoring of ecological environment and human activities, and strengthening supervision and investigation of ecological damage issues

Strengthen the supervision and punishment of various illegal and irregular behaviors.

The management of wildlife in the whole region will be included in the assessment of the ecological environment protection target responsibility system.

Survey of cultivated and preserved forest and grass germplasm resources, and strengthen forest and grass germplasm resources that adapt to climate change

Resource preservation library construction. Accelerate the process of legalization of ecological protection compensation, formulate and implement

Ecological protection compensation method, and continuously improve the ecological protection compensation mechanism. (Municipal Planning Resources

The Municipal Bureau of Ecology and Environment, the Municipal Development and Reform Commission shall be responsible for the division of responsibilities)

16. Improve disaster warning, prevention and governance capabilities. Strengthen the prevention and control of animal and plant diseases at the grassroots level.

Build a prevention and control system, improve the monitoring network of major crop diseases and insect pests, promote biological control,

Green prevention and control technologies such as ecological regulation. Conduct a census of alien invasive species in forests and grasslands. Strictly control agricultural

Crop and forest and grass seeds and seedlings, aquatic seedlings, etc. have been introduced for approval and quarantine supervision, and the

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Investigation, monitoring, early warning, control, assessment and removal of invasive alien species in key areas and ecological restoration. Support and encourage insurance institutions to provide docking services and develop Insurance products for disasters such as forest and grass fires, low temperature ice and snow, biological pests and diseases, and biological invasions. Improve post-disaster recovery capabilities. Strengthen the application of satellite and lidar data in sand and dust monitoring. Use to improve the accuracy of sandstorm disaster warning, prediction and forecast. (Municipal Planning and Resources Bureau, Municipal Agriculture and Rural Affairs Commission, Municipal Ecological Environment Bureau, Tianjin Customs, Municipal Meteorological Bureau, National Financial The Tianjin Supervision Bureau of the State Administration of Supervision and Administration and the Municipal Finance Bureau are responsible for the division of responsibilities)

17. Implement the protection and restoration of important ecosystems. Strengthen the protection of national important wetlands and The city has strengthened the protection and restoration of important wetlands, including Qilihai, Dahuangbao, and Beida. The protection and restoration of the four wetland nature reserves in Gang, Tuanbo and Dahuangbao were promoted Nationally Important Wetlands. By 2025, the wetland protection rate will reach 60%. By 2035, The wetland protection rate is no less than 60%. Continue to carry out ecological restoration of abandoned mines and strengthen forest protection. Protect and rebuild the natural ecology of the mountains and enhance the ecological conservation function of the northern mountainous areas. Forest resource protection and management system, strictly implement forest land quota management and forest felling quota management management, and effectively strengthen forest cultivation and management and transformation of inefficient forests. The area has reached more than 5.5 million cubic meters, and the forest coverage rate has reached 13.6%. (City Planning The Resources Bureau and the Municipal Ecological Environment Bureau are responsible according to their respective duties)

18. Strengthen biodiversity conservation and implement major biodiversity conservation projects. Priority areas for biodiversity conservation, nature reserves, ecological protection red lines and other important Focus on ecological space and carry out biodiversity surveys and observations. Gradually build a biodiversity Protect the network, focus on protecting bird migration corridors and oyster reef protection areas, and increase the protection of relict gulls, We will strengthen the protection and rescue of rare and endangered wild animals and plants such as the Oriental White Stork and the Phellodendron amurense, and crack down on

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Illegal and criminal acts that damage wild animal and plant resources. Strengthen the protection and management of biological genetic resources.

Strengthen the protection and utilization of agricultural germplasm resources. Continuously improve the policy on wildlife protection.

(Municipal Ecological Environment Bureau, Municipal Planning and Resources Bureau, Municipal Agriculture and Rural Affairs Commission,

The Municipal Public Security Bureau is responsible according to the division of responsibilities)

10. Ocean and Coastal Zone

19. Improve the marine disaster observation, early warning and assessment system. Strengthen the long-term

Conduct long-term monitoring surveys, impact assessments and forecasts, strengthen ground subsidence and embankment elevation monitoring,

Improve the early warning and comprehensive risk assessment capabilities of extreme high sea levels. Continue to do a good job in sea ice,

Early warning and monitoring of marine disasters such as storm surges, strengthening emergency monitoring of marine disasters during the flood season, and promoting

Sharing of marine disaster information. Implementing operational marine ecological early warning monitoring, focusing on

Early warning monitoring of red tides and emergency monitoring of red tides, green algae and jellyfish along the coast of Tianjin.

Marine meteorological monitoring, analysis, forecast and early warning system to improve the marine meteorological conditions in coastal and offshore areas

Disaster and extreme weather event monitoring, forecasting and early warning capabilities. Build marine meteorological risk management

System, and carry out risk assessment for meteorological disasters such as strong winds and sea fog at sea. (Municipal Planning

The Resources Bureau, Municipal Meteorological Bureau, and Municipal Water Affairs Bureau are responsible for their respective duties.

20. Improve the disaster prevention and mitigation capabilities of coastal areas and coastal areas.

The magnitude and marine disaster warning will be improved to improve the disaster prevention standards of coastal infrastructure.

The tide-proof seawalls in the district should be built up to standard, and new seawalls should be built in the land reclamation areas in due course.

Improve the ability of coastline protection and resistance to marine disasters. Strengthen storms during seasonal high sea level periods

To ensure the safety of coastal areas from floods.

The moisture-proof standard is achieved to reach once in 100-200 years. Strictly control ground subsidence and build a comprehensive

A three-dimensional land subsidence monitoring network will be established. By 2035, the land subsidence in the plain area of the city will be

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Comprehensive control. (The Municipal Water Affairs Bureau, Municipal Meteorological Bureau, and Municipal Planning and Resources Bureau are responsible for the division of responsibilities.

responsibility)

21. Strengthen the ecological protection and restoration of coastal waters. Implement the “Blue Bay” renovation and restoration.

Strictly implement the marine nature reserve and ecological protection

Protect the red line supervision, strengthen the classification control and optimization of coastal zone utilization, in addition to major national strategies

By 2025, the natural coastline reserves shall be no less than

18 km, the proportion of the marine ecological protection red line to the jurisdiction area remains above 10%

By 2030, the amount of natural coastline will not decrease.

Coordinate and deploy to promote the restoration of estuary and bay ecosystems. Strictly implement the summer fishing moratorium in the ocean

Improve the marine biodiversity survey,

Monitoring, evaluation and protection system. (Municipal Planning and Resources Bureau, Municipal Water Affairs Bureau, Municipal Agriculture and Rural Affairs Bureau

The village committee and the Municipal Ecological Environment Bureau are responsible according to their respective duties)

22. Continue to improve the quality of the marine ecological environment. Strengthen the “one river, one policy” approach for rivers flowing into the sea.

Governance, carry out in-depth investigation and rectification of sewage outlets into the sea, and strengthen pollution prevention and control of marine aquaculture.

Strengthen the prevention and control of port ship pollution and strictly implement the reception, transfer and disposal of ship pollutants

Joint supervision system. Promote the protection and construction of beautiful bays with "one bay, one policy".

In 2017, the proportion of coastal waters with excellent water quality (Class I and Class II) reached 72%.

Environmental Protection Bureau, Municipal Planning and Resources Bureau, Municipal Transportation Commission, Tianjin Maritime Safety Administration, Municipal Urban Management Bureau

The Municipal Commission, Municipal Water Affairs Bureau, and Municipal Agriculture and Rural Affairs Commission are responsible for their respective duties.

V. Improving the ability of economic and social systems to adapt to climate change

11. Infrastructure and Human Living Environment

23. Strengthen climate risk management of urban and rural infrastructure.

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Upgrade and transform existing facilities to fill the gaps in infrastructure and public service facilities and promote green development of existing facilities.

Improve emergency water supply, emergency power supply, emergency broadcast communication, emergency garbage collection and disposal.

garbage and sewage treatment systems to ensure that lifeline facilities operate well in emergencies, and

Reserve safety reserves. Adjust the elevation of buildings in coastal areas and

Design requirements for relevant protective facilities. Integrate meteorological services into the construction of the "city brain"

Improve the meteorological services to ensure urban water, electricity, gas, heating, flood control and drainage, and transportation

(Municipal Housing and Urban-Rural Development Committee, Municipal Urban Management Committee, Municipal Agriculture and Rural Affairs Committee,

Municipal Water Affairs Bureau, Municipal Development and Reform Commission, Municipal Industry and Information Technology Bureau, Municipal Planning and Resources Bureau,

The Municipal Meteorological Bureau is responsible according to the division of responsibilities)

24. Improve the service functions of urban ecosystems.

Repair, increase the proportion of urban "blue and green space", open up urban ventilation corridors, and alleviate urban

Improve the urban "heat island effect" and improve the urban living environment.

Green ecological barrier zone construction, accelerate the promotion of afforestation, water system connectivity and ecological

Through restoration and other projects, an ecological corridor will be constructed running through the north and south of Tianjin. By 2035,

The area of "blue-green space" in the barrier zone shall not be less than 70%. Scientific planning and layout of the city

Green rings, green corridors, green wedges and greenways coordinate green space construction and urban space development to create a safe and

Comfortable, convenient, practical and beautiful green space. Promote the "Botanical Garden Chain" in an orderly manner

Project construction, and strive to promote the construction of pocket parks. (Municipal Planning and Resources Bureau, Municipal Urban Management Bureau

The management committee is responsible according to the division of responsibilities)

25. Strengthen the construction of urban waterlogging drainage capacity. Focus on "Tianjin City" and "Binhai

The city and other urbanized areas are relatively independent units, and the overall promotion of urban drainage in the city

System construction. Implement the renovation of drainage facilities in areas prone to waterlogging and tunnels prone to waterlogging, and accelerate the renovation of old and dilapidated

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Update and upgrade pipeline network facilities, promote the management of combined rainwater and sewage systems, and strengthen the management of secondary rivers

By 2025, the city's drainage and flood control capabilities will be significantly improved.

There will be no water accumulation on the main road network in the city core area 2 hours after the heavy rain stops, and no water accumulation on the main road network in the city core area 5 hours after the heavy rain stops.

All accumulated water will be drained. By 2035, the city will basically meet the requirements of resilient urban construction.

Urban drainage and flood control engineering system integrating the concept of sponge city, urban carrier function and drainage

The ability to prevent waterlogging has been significantly improved. Promote the construction of sponge cities across the region. By 2025, 45%

The above built-up areas meet the requirements of sponge cities. (Municipal Water Affairs Bureau, Municipal Housing and Urban-Rural Development Bureau

The committee is responsible for the division of responsibilities)

12. Agriculture and Food Security

26. Optimize the utilization pattern of agricultural climate resources.

Refine the agricultural climate zoning and the major agricultural meteorological disaster risk zoning.

Under the premise of full production, we should adhere to the principle of water-based production, appropriately adjust and optimize the layout of agricultural regions and agricultural

Continue to improve farmland infrastructure construction and expand high-standard farmland construction

Under the premise of completing the national arable land protection and food security assessment tasks, reduce high

Water-consuming crops should be planted and rain-fed agriculture should be developed.

Introduce and cultivate vegetables suitable for the city's phenology

The conditions are good for the production of crops with strong stress resistance and high light efficiency, optimizing the variety structure and maintaining the crops

Higher multiple cropping index. (Municipal Agriculture and Rural Affairs Commission, Municipal Meteorological Bureau, Municipal Development and Reform Commission

Division of responsibilities

27. Strengthen the construction of agricultural disaster response and mitigation capabilities. Focus on food security and important agricultural products

Ensure the supply of agricultural products, strengthen the basic capacity of urban agricultural meteorology, and strengthen agricultural meteorology at key farming seasons.

Monitoring and forecasting of meteorological disasters, and improving the ability of agriculture to cope with extreme meteorological disasters.

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Develop emergency plans for natural disasters and strengthen the storage of agricultural disaster prevention and mitigation materials such as famine-stricken seeds.

Agricultural adaptation to climate change technology innovation, demonstration and promotion of climate change adaptation technology according to local conditions

Relying on high-quality farmer training and expert guidance from industry technology system, we continue to provide

Provide technical guidance and training on disaster prevention, mitigation and adaptation for producers and operators.

Agricultural disaster insurance such as climate quality insurance will improve the agricultural meteorological risk sharing mechanism.

Agriculture and Rural Affairs Commission, Municipal Meteorological Bureau, Tianjin Supervision Bureau of the State Financial Supervision and Administration Bureau, Municipal

The Finance Bureau is responsible according to the division of responsibilities)

28. Enhance the climate resilience of agricultural ecosystems. Strengthen farmland construction and management.

Promote conservation tillage and intercropping according to local conditions.

The technical model of water-fertilizer integration, soil testing and formula fertilization, and integrated prevention and control of pests and diseases mainly

The utilization rate of chemical fertilizers for crops and the utilization rate of pesticides for major crops have reached national target requirements.

Vigorously develop efficient water-saving irrigation, improve irrigation facilities such as reservoirs and canals,

Efforts should be made to improve the drought and flood resistance of agricultural production.

and identification and evaluation, build a crop germplasm resource bank in our city, and strengthen the control of agricultural invasive species

Species monitoring and prevention, and protection of agricultural biodiversity.

The bureau is responsible according to the division of responsibilities)

13. Health and Public Health

29. Conduct climate change health risk and adaptation capacity assessment.

cross-cutting and deep integration, establish joint research and cross-innovation mechanisms, and carry out climate change research

Monitoring, investigation, statistics, and risk assessment of human health impacts, effectively clarifying and identifying

Climate change health risks and vulnerable groups. Based on the results of climate change health risk assessment,

Taking into full consideration the climate characteristics of the region and the health risk exposure level of vulnerable groups,

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Assess the capacity of health systems and key vulnerable groups to adapt to climate change and develop adaptation capacity

(The Municipal Health Commission and the Municipal Meteorological Bureau are responsible for the division of responsibilities)

30. Strengthen the monitoring, early warning, and prevention and control of climate-sensitive diseases. Strengthen the prevention and control of influenza and chickenpox.

Monitoring of seasonal epidemic respiratory infectious diseases such as

The system is designed to treat plague, avian influenza and other zoonotic diseases and COVID-19 that are susceptible to climate factors.

Focus on monitoring key infectious diseases such as cholera, dengue fever, and malaria.

Monitoring of chronic non-communicable diseases such as pneumonia and respiratory diseases and their correlation with climate.

Innovate and improve the new model of "meteorology + health care + service" and explore the establishment of a new model for climate change

Monitoring, early warning and public information release mechanisms for sensitive diseases, and improving the prevention and control of stroke and high temperature

Strictly implement extreme weather and climate warnings such as high temperature and heat waves.

Labor protection standards for operations under incident conditions. (Municipal Health Commission, Municipal Meteorological Bureau

Division of responsibilities)

31. Enhance the climate resilience of the medical and health system. Innovate the medical and prevention collaborative work mechanism.

Provide first aid, treatment and nursing care for climate-sensitive diseases in a graded, stratified and diverted manner

Improve emergency response plans for public health emergencies and medical and health rescue for emergencies

Support emergency response plans to prepare for climate-sensitive diseases and health risks under extreme weather and climate events

Strengthen emergency drills and personnel training, and improve public health response.

Emergency supplies and medical supplies reserve mechanism, build a city-wide emergency response system for public health emergencies

Establish a command platform to improve emergency response to public health emergencies under extreme weather and climate events

Improve the psychological preparedness of extreme weather and climate events.

Crisis intervention mechanism, strengthen the construction of psychological assistance hotlines at the municipal and district levels, and improve psychological health

Kanghe mental health service capacity level. (Municipal Health Commission)

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(XIV) Sensitive secondary and tertiary industries

32. Improve the meteorological service guarantee capability and build the "intelligent forecast + meteorological service" business system, develop commercial meteorological services, and promote smart, precise, open, integrated, and inclusive

Strengthen the aggregation of industry big data and meteorological data,

Correlation analysis and application, establish meteorological impact service models and precise forecasts for different industries and disaster types.

Refine the risk warning model and develop it for multiple fields covering agriculture, industry and service industry

Meteorological service products improve electricity, transportation, ports, marine engineering, tourism, and finance

The ability of sensitive secondary and tertiary industries such as insurance to cope with extreme weather and climate events and their secondary disasters

(Municipal Meteorological Bureau)

33. Improve the climate resilience of the energy sector. Promote climate-friendly technologies such as offshore wind and solar energy.

Promote the establishment of large-scale solar energy, wind energy and other climate resource development

Establish a climate feasibility study and demonstration system for development and utilization projects and a meteorological disaster risk assessment system.

In response to extreme weather and climate events such as high temperature and freezing, we will conduct

Optimize energy supply based on climate resources and energy demand

structure, rationally arrange peak-shaving power sources, and strengthen the construction of power emergency peak-shaving capacity.

Protection and emergency dispatch of power transmission and distribution systems under extreme weather and climate events, and strengthening of power equipment monitoring

Improve the emergency response plan system, promote energy storage, smart grid and digital

Improve the prediction, early warning, defense and response of power infrastructure safety risks through the application of technologies such as

and rapid recovery capabilities.

Disaster weather monitoring, early warning and emergency response plans to ensure natural gas supply in Beijing-Tianjin-Hebei and North China

Supply and safety. (Municipal Meteorological Bureau, Municipal Bureau of Industry and Information Technology, Tianjin Maritime Safety Administration, Municipal

Development and Reform Commission, Municipal Urban Management Commission, Municipal Transportation Commission, Municipal Emergency Management Bureau, Municipal

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The power company is responsible according to the division of responsibilities)

34. Strengthen transportation disaster prevention and emergency response. Strengthen the construction of resilient transportation infrastructure.

Integrate climate change adaptation requirements into the entire process of transportation infrastructure construction,

Strengthen disaster prevention and control projects for transportation facilities and build a resilient and reliable transportation safety system.

The transportation emergency plan system is linked to the natural disaster emergency plan system.

Establish a professional emergency rescue team for emergency response to highway emergencies and enhance water search and rescue capabilities.

Improve the city-level highway traffic emergency equipment and material reserve system, and supervise the improvement of district-level highway traffic emergency equipment and material reserve system.

Establish emergency equipment and material reserve system to improve emergency rescue of municipal highway emergency support base

(The Municipal Transportation Commission, the Municipal Emergency Management Bureau, and the Tianjin Maritime Safety Administration shall divide responsibilities and

Responsible)

35. Prevent climate-related financial risks. Promote financial institutions to strengthen environmental information disclosure.

Build capacity for environmental information disclosure and encourage qualified financial institutions to disclose environmental information.

Listed companies and bond-issuing companies included in the list of companies that disclose environmental information in accordance with the law shall disclose their environmental information in accordance with the law.

Encourage financial institutions to disclose information related to high-carbon asset exposure. Actively explore

The feasibility of conducting climate risk sensitivity stress testing on corporate banks within the jurisdiction is sought.

Encourage and guide financial institutions to formulate green finance and transition finance development plans, and improve relevant mechanisms.

Actively promote financial institutions to innovate financial products and serve the development of green and low-carbon enterprises.

(The People's Bank of China Tianjin Branch, the Municipal Financial Bureau, and the Municipal Ecological Environment Bureau shall be responsible for

Worker responsible)

36. Develop climate-adaptive tourism. Carry out meteorological disaster risk assessment in key tourist attractions.

Risk assessment and monitoring and early warning, strengthen the protection of scenic spots and historical sites threatened by climate change,

Protection of endangered cultural and natural heritage. Combined with the city's four-season climate and tourism resources

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Characteristics, develop sightseeing, science popularization, and vacation industries that are attractive to Chinese and foreign tourists according to local conditions

Improve emergency response to extreme weather and climate events at tourist destinations and tourist safety management

Establish and improve the classification and analysis of special landscape appreciation index and tourism meteorological index of tourist attractions.

Level indicators, improve the ecological tourism meteorological service guarantee capabilities. (Municipal Culture and Tourism Bureau, Municipal Meteorological Bureau

The bureau is responsible according to the division of responsibilities)

VI. Strengthening the Climate Resilience of National Space

15. Optimizing the regional pattern of adaptation to climate change

37. Build a national space that adapts to climate change. Implement the national space development strategy.

Incorporate the ability to respond to climate change into national land space planning and build a “three zones, two belts,

The ecological space pattern of "barrier" should be strictly observed.

Spatial control boundaries such as farmland, ecological protection red lines, and urban development boundaries are planned in a coordinated manner

Urban construction, industrial development, ecological conservation, infrastructure and public services, reasonable regulation

Plan urban layout and functions to minimize the climate risks faced by cities.

The city's climate impact, disaster comprehensive risk assessment information and land space information are integrated to enrich

Urban "one map" system. In the preparation of national land space planning and related special planning,

Fully consider the impact of sea level rise and optimize the spatial layout of the coastal zone.

Planning and climate feasibility analysis of key engineering projects. (Municipal Planning and Resources Bureau, Municipal Ecological

The Environmental Bureau and the Municipal Meteorological Bureau are responsible for the division of responsibilities)

38. Deepen the coordinated adaptation in the Beijing-Tianjin-Hebei region. Upgrade the meteorological emergency video conference system.

Improve the cross-regional meteorological disaster risk assessment capabilities and strengthen the cross-regional meteorological disaster risk assessment capabilities in the Beijing-Tianjin-Hebei region.

Regional emergency coordination and linkage, strengthen cross-regional emergency drills. Promote the improvement of the Beijing-Tianjin-Hebei region

The emergency command center has a regular consultation mechanism to strengthen cross-border cooperation with neighboring districts and counties in Beijing and Hebei and relevant departments.

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Regional resource sharing and joint guarantee will enhance cross-regional disaster emergency response capabilities.

Actively participate in the construction of joint weather modification capabilities in Beijing, Tianjin and Hebei, and provide water resources for the water conservancy projects in the capital area.

Resource security, national food security, ecological security, and emergency support services are provided efficiently and effectively

(The Municipal Emergency Management Bureau, Municipal Meteorological Bureau, and Municipal Water Affairs Bureau are responsible for the division of responsibilities.

responsibility)

16. Explore differentiated climate adaptation paths

39. Strengthen the ability to respond to climate risks in key areas. Continue to increase forest

Strengthen forest fire prevention efforts, strengthen forest fire risk consultation and assessment in extreme weather conditions, and issue forest fire

Risk warning and reminder, strengthen forest fire prevention informationization and high-tech fire prevention and extinguishing construction.

Prevention is the priority, and avoidance and control are combined to strengthen the prevention and control of mountain torrents, mud-rock flows, and landslides in the northern mountainous areas.

The prevention and control of sudden disasters induced by weather such as slopes, and the establishment and improvement of meteorological level forecasting for geological disasters

Strengthen coastal areas to prevent marine meteorological disasters such as storm surges, sea ice and ground subsidence.

Comprehensive prevention and control. Strengthen the evolution and impact of the "Tianjin City" and "Binhai City" twin-city heat island phenomenon

Impact research will be carried out to strengthen the climate change risk assessment and adaptation capabilities of "Tianjin City" and "Binhai City".

Promote the "one-network management of urban operations" and strengthen municipal infrastructure, environment, and transportation.

The interconnection of various urban operation systems such as transportation and transportation can improve rapid response and efficient linkage disposal

Promote the connection between government service platforms, community sensing facilities and home terminals, and develop

Intelligent early warning, emergency rescue and relief, post-disaster psychological recovery and other community services.

Planning and Resources Bureau, Municipal Emergency Management Bureau, Municipal Meteorological Bureau, Municipal Water Affairs Bureau, Municipal Urban Management Bureau

The Municipal Commission of Development and Reform, the Municipal Transportation Commission, and the Municipal Health Commission shall be responsible for the division of responsibilities.

responsibility)

40. Explore the implementation of climate adaptation pilot projects. Explore the adaptation of mega-coastal cities.

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Climate change development path, support qualified districts to apply for national climate-adaptive city construction

Set up pilot projects to encourage key areas, key industries and sensitive regions to explore and carry out climate adaptation

Innovative practices will drive the overall improvement of the city's ability to adapt to climate change.

The construction of the National Meteorological Science and Technology Park will build an international smart city meteorological demonstration zone and enhance Tianjin

The meteorological guarantee service capacity of socialist modern metropolis. Support Binhai New Area in low-carbon

Take the lead in innovation in project investment and financing, and promote innovation in climate investment and financing models and mechanisms.

The Ecological Environment Bureau and the Municipal Meteorological Bureau are responsible according to their respective duties)

VII. Safeguard Measures

(XVII) Strengthening organization and implementation

Strictly implement national deployment and establish a municipal-level organization and implementation team for climate change adaptation.

Communication and coordination mechanism to coordinate and promote climate change adaptation work.

work, refine implementation measures, consolidate work tasks, open up information channels, and form a

The people's governments of all districts take measures according to local conditions and take positive actions, setting high standards and

Quality promotes the implementation of various tasks. Establish an evaluation mechanism for the effectiveness of adaptation to climate change.

Carry out assessments of action plans for adaptation to climate change in a timely manner and implement them according to the assessment results and the situation.

Adjustments and optimizations are needed. (Relevant departments are responsible for their respective duties)

18. Increase capital investment

Strengthen the coordination of financial resources, actively seek central financial support, and increase local funding

Raise funds, optimize the structure of fiscal expenditure, and strengthen the adaptation to key areas and regions.

Strengthen the protection of climate change work. Give full play to the incentive role of green finance evaluation and guide

Financial institutions invest in climate adaptation projects. Guide market-oriented and diversified investment, combine fund main

Key investment areas: support government-led funds to increase investment in climate change adaptation,

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Guide social capital to strengthen its support function in related fields. Strengthen the government-bank cooperation in climate investment and financing

Connecting financial institutions with enterprises to accurately match the financing needs of enterprises.

Financial Bureau, People's Bank of China Tianjin Branch, Municipal Development and Reform Commission, Municipal Ecological Environment

The bureau is responsible according to the division of responsibilities)

(19) Strengthen scientific and technological support

Strengthen basic research on addressing climate change and actively promote

Key and forward-looking technologies are being tackled.

National and municipal key scientific research projects, guide and promote technology innovation centers, industrial innovation

Alliances, key enterprise laboratories and other platforms strengthen technical research on addressing climate change.

Promote the construction of an innovation system for adapting to climate change and enhance innovation capabilities.

Transformation of advanced technological achievements in changing fields, promote the connection between industry, academia and research, and help adapt to technology promotion

(The Municipal Science and Technology Bureau, Municipal Meteorological Bureau, and Municipal Ecological Environment Bureau are responsible for the division of responsibilities)

20. Promoting the participation of all people

Take advantage of World Meteorological Day, World Earth Day, World Environment Day, World Water Day,

National Disaster Reduction Day, National Low Carbon Day and other important nodes to popularize climate change knowledge and

Raise the public's awareness of climate change. Widely mobilize enterprises, communities, associations,

Citizens actively participate in climate change adaptation work and promote the diversification of adaptation action subjects.

Support the construction of science popularization bases in nature reserves based on ecological environmental observation stations and tourist sites.

(Municipal Ecological Environment Bureau, Municipal Meteorological Bureau, Municipal Science and Technology Bureau, Municipal Planning and Resources Bureau shall be responsible for the following duties:

Worker responsible)